

Module-4

Protection of Fault Secured IP Core using Digital Signature based Watermark

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Introduction

- Reusable IP cores—specifically DSP and multimedia cores—are part of almost every SoC in CE products.
 - For example, JPEG compression process used in image and video compression of camera systems, discrete cosine transform (DCT) filter used in lossy compression of audio and image files, FFT used in digital video broadcasting, finite impulse response (FIR) filters used in audio devices such as headsets etc., and infinite impulse response (IIR) filters used in audio processors for equalization of speakers.
 - Current design challenges in the domain of consumer electronics include greater complexity and stiff time to market pressure.
 - Reusable IP cores, often delivered by a third-party vendor, serve as a viable solution for maximizing the design productivity while minimizing design time for CE products.
 - The global semiconductor supply chain for SoC design is highly vulnerable to piracy, IP forgery or fraudulent ownership abuse.
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Background on different IP Core Threats and Countermeasures

❑ IP Piracy and counterfeiting:

- A well-known threat to DSP based IP cores and chip sets are direct piracy, where fraudulent means can result in direct theft or duplication of the IP design for illegal distribution.
- After obtaining the design, the adversary can even falsely claim its ownership.

❑ Several solutions have been proposed

1. For example, authors in (Koushanfar et al., 2005; Sengupta and Bhadauria, 2016; Sengupta et al., 2018; Sengupta and Roy, 2017; Sengupta and Roy, 2018; Sengupta and Mohanty, 2019) embed a robust watermark in the IP core design process.
2. (Yu and Zhu, 2011) presents a watermarking method for soft IP core protection in which a special watermark module for copyright detection is added.
3. A watermarking approach based on two dimensional chaotic mapping model is proposed in (Liang et al., 2015).
4. Approach (Roy and Sengupta, 2017) proposes a **symmetrical protection** method, wherein, both seller watermark and buyer fingerprint are embedded in an IP design.
5. Non-watermark type of protection mechanisms against IP piracy in the literature. For instance, authors in (Alkabani and Koushanfar, 2007) propose the concept of **IP metering**.

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❑ **Reverse Engineering and Trojan insertion:**

- Reverse engineering (RE), employed to extract the relevant information such as specification and architecture of an IP core design.
- Once the required information is available, the adversary can duplicate the design by reconstructing the controller, or introduce malicious Trojan in the original design.

❑ **Several solutions have been proposed**

1. Obfuscation is a frequently employed approach to prevent RE attacks (Chakraborty and Bhunia, 2009).
 2. For example, high level transformations (HLTs) based structural obfuscation is presented in (Lao and Parhi, 2015) to protect DSP cores against RE threats.
 3. Further, IP core locking based functional obfuscation used in (Sengupta et al., 2019) and a dual modular redundant design based HLS (Sengupta et al., 2010) methodology for Trojan detection are also presented in (Sengupta et al., 2017).
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Brief Description of Fault Security for IP cores

- Integrated circuit (IC) packages contain some trace amounts of uranium and thorium impurities which emit alpha particles.
 - Ionizing radiation from these alpha particles causes single-event upset (SEU), leading to a single-event transient.
 - Due to the presence of α -particles in IC packages, fault secured IP cores are becoming popular for SoC applications.
 - An SEU can result in functional failures such as producing an erroneous output.
 - The increase in density per unit area in a chip—due to the massive technology scaling—is increasing, rate of occurrence of transient faults.
 - Moreover, the advent of faster (high frequency) devices has resulted in multi-cycle transient faults that last more than one control step.
 - Transient fault secured IP core implies, the design is secured against faults that have maximum transient fault strength, i.e., worst case impact of a transient fault.
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Threat Model

- The threat models for digital signature based watermarking are protection of fault secured DSP based IP cores from the following:
- (i) IP/IC piracy
 - (ii) IP/IC counterfeiting and cloning
 - (iii) Fraudulent claim of ownership
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Selected Contemporary Approaches

1. Similar Prior Approaches

- So far in the literature, few watermarking techniques have been proposed for protection of DSP based IP cores (Sengupta and Bhadauria, 2016; Sengupta et al., 2018; Yu and Zhu, 2011; Roy and Sengupta, 2017).
 - **Watermarking is a passive mode of protection**, wherein a hidden mark is implanted in the design to protect reusable IP cores against ownership abuse.
 - This is non-trivial, since the choice of design abstraction level affects the design cost overhead, security and implementation complexity. Watermarking has been proposed at lower levels of abstraction such as gate level or register transfer level in (Nie et al., 2013; Gal and Bossuet, 2012; Ziener and Teich, 2008).
 - For instance, a hierarchical watermarking scheme for protection of FPGA IP cores is presented in (Nie et al., 2013), wherein the netlist and bit stream of an IP design are used to insert a vendor watermark.
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Selected Contemporary Approaches

- Moreover, authors in (Ziener and Teich, 2008) proposed a power watermarking technique that detects signature at power supply pins of FPGA.
 - Further, authors in (Koushanfar et al., 2005) employ a binary encoding scheme to implant the watermarking constraints in the IP core design, while authors in (Sengupta and Bhadauria, 2016; Sengupta et al., 2018) use a multi variable encoding scheme.
 - (Sengupta et al., 2018) embeds vendor signature at three phases of HLS design.
 - However digital signature based watermarking approach provides more definite proof of authorship (stronger security) using crypto-based signature process and incurs lower overhead than (Koushanfar et al., 2005; Sengupta and Bhadauria, 2016; Sengupta et al., 2018).
 - However, in the context of fault secured DSP cores, implanting the signature at lower abstraction level increases design complexity, does not provide comprehensive security and enhances chances of design overhead. Thus above approaches are not capable of handling fault secured DSP cores, unlike presented approach.
 - **Further, there exists no work that proposes protection of fault secured DSP cores using robust watermark.**
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Forensic Detective Control of Fault secured IP cores using Digital Signature based Watermark

1. Input/Output

Inputs: (a) Control data flow graph (CDFG) of a DSP core (b) Strength of transient fault (K_c) (c) Loop unrolling factor (UF) (d) Module library and (e) User specified hardware resources set $(X) = N(R_1), N(R_2), N(R_D)$, where $N(R_D)$ is the number of hardware resources of type R_D ;

Output: Fault secured watermarked DSP design.

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2. Fault secured DSP IP cores Protection Framework

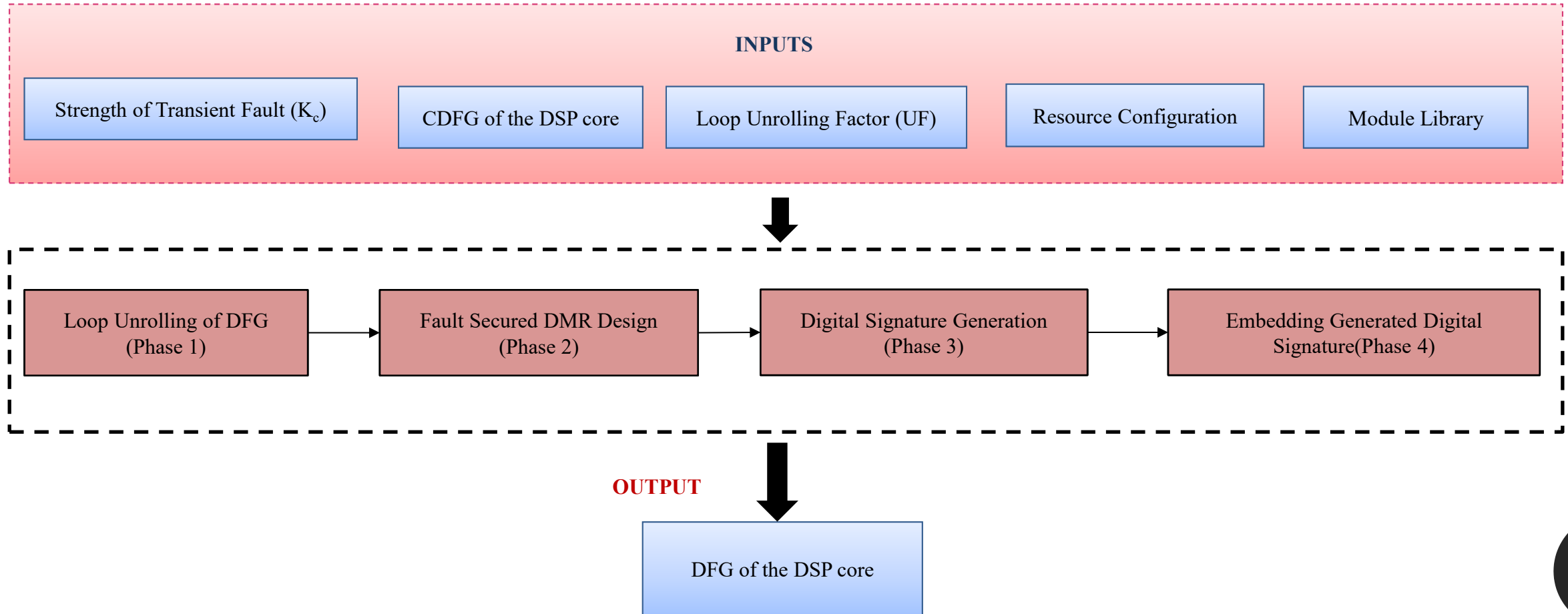


Fig. Overview of digital signature embedding approach in terms of different phases

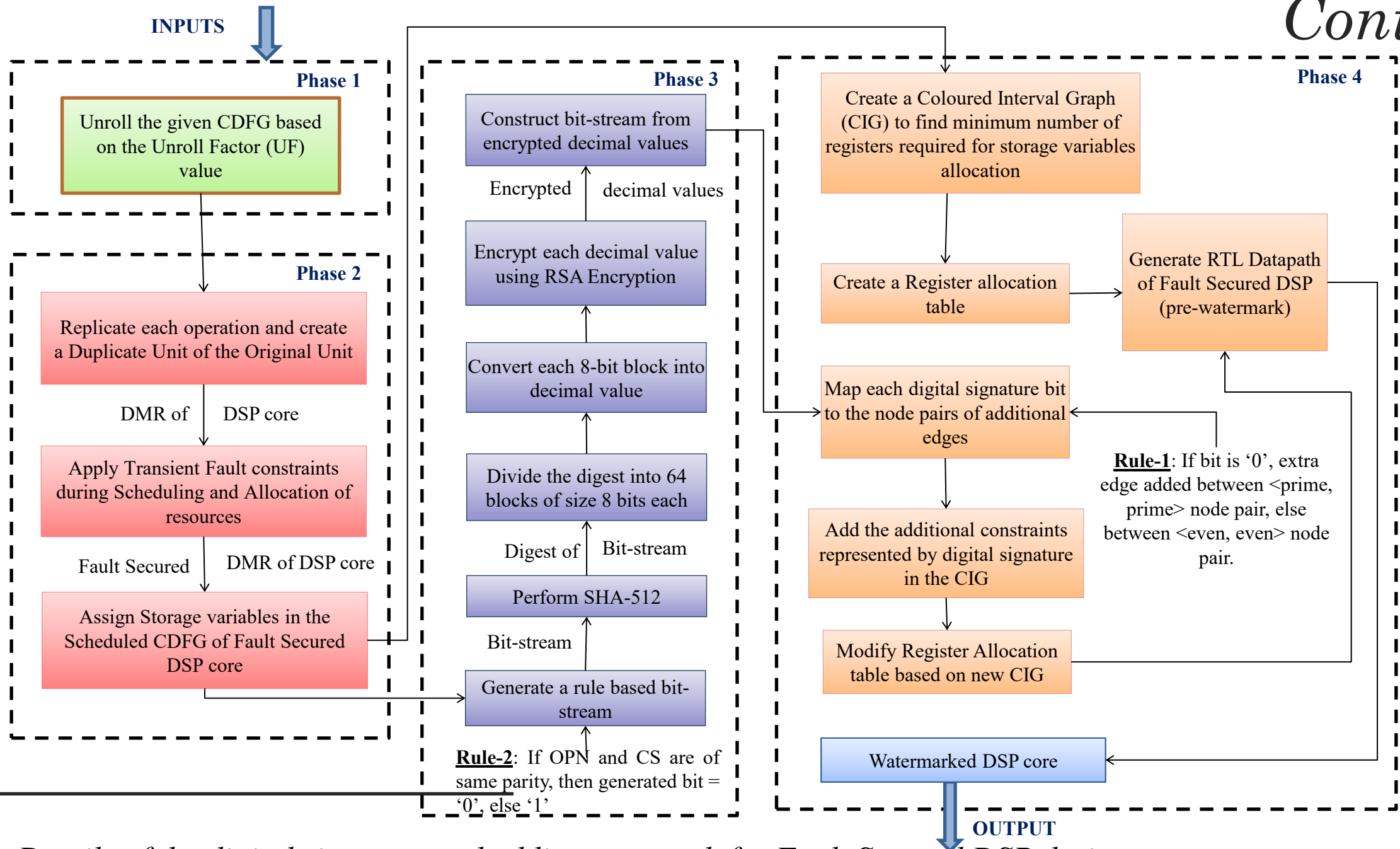
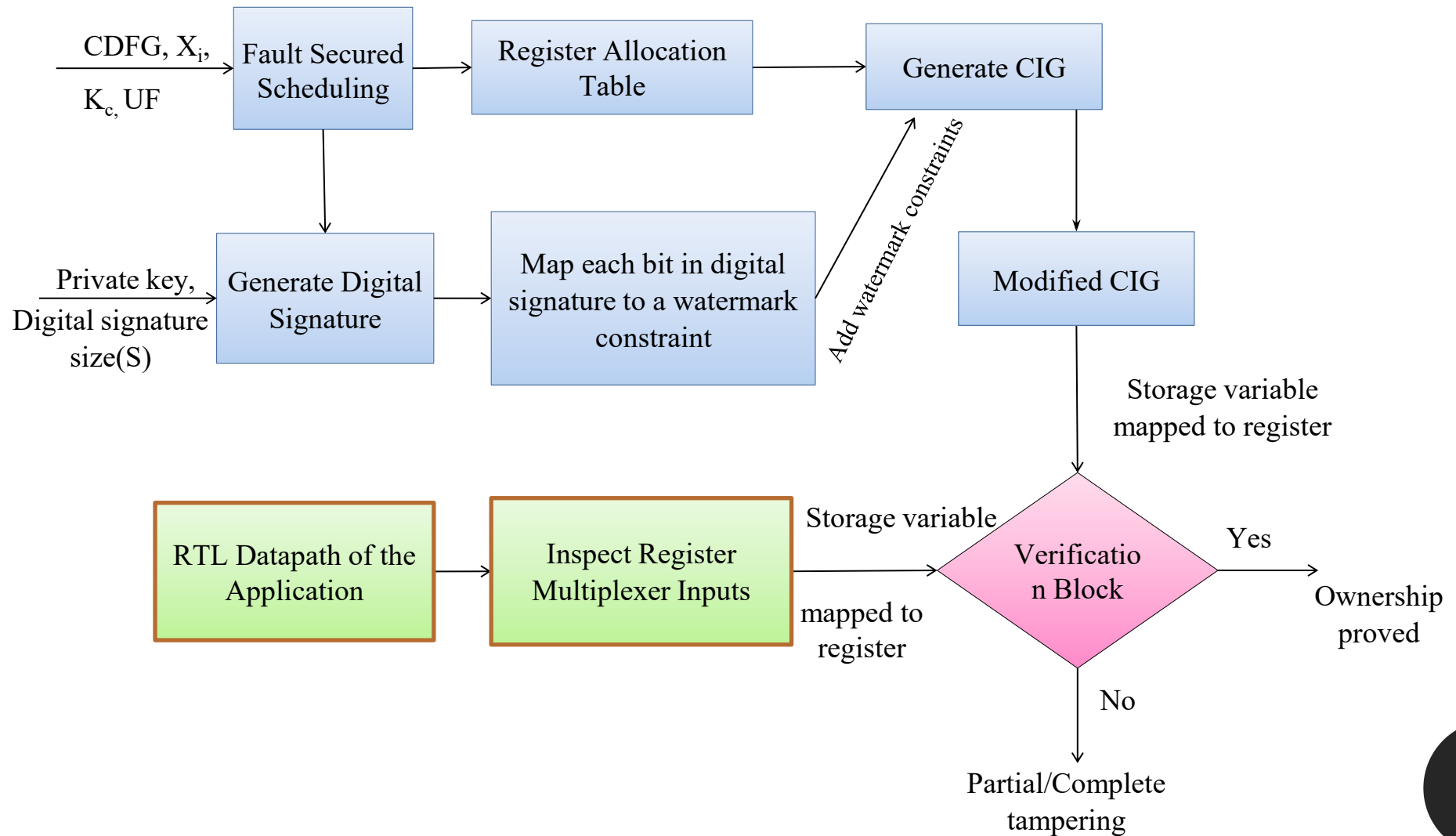


Fig. Details of the digital signature embedding approach for Fault Secured DSP design

Signature Detection process



- ❑ To resolve the conflict of false claim of ownership or detect IP piracy/counterfeiting, the vendor/owner signature can be extracted from the fault secured IP core as shown here:

Fig. *Process of detecting digital signature for fault secured IP cores*

Case Study on Fault Secured Finite Impulse Response (FIR) Filter

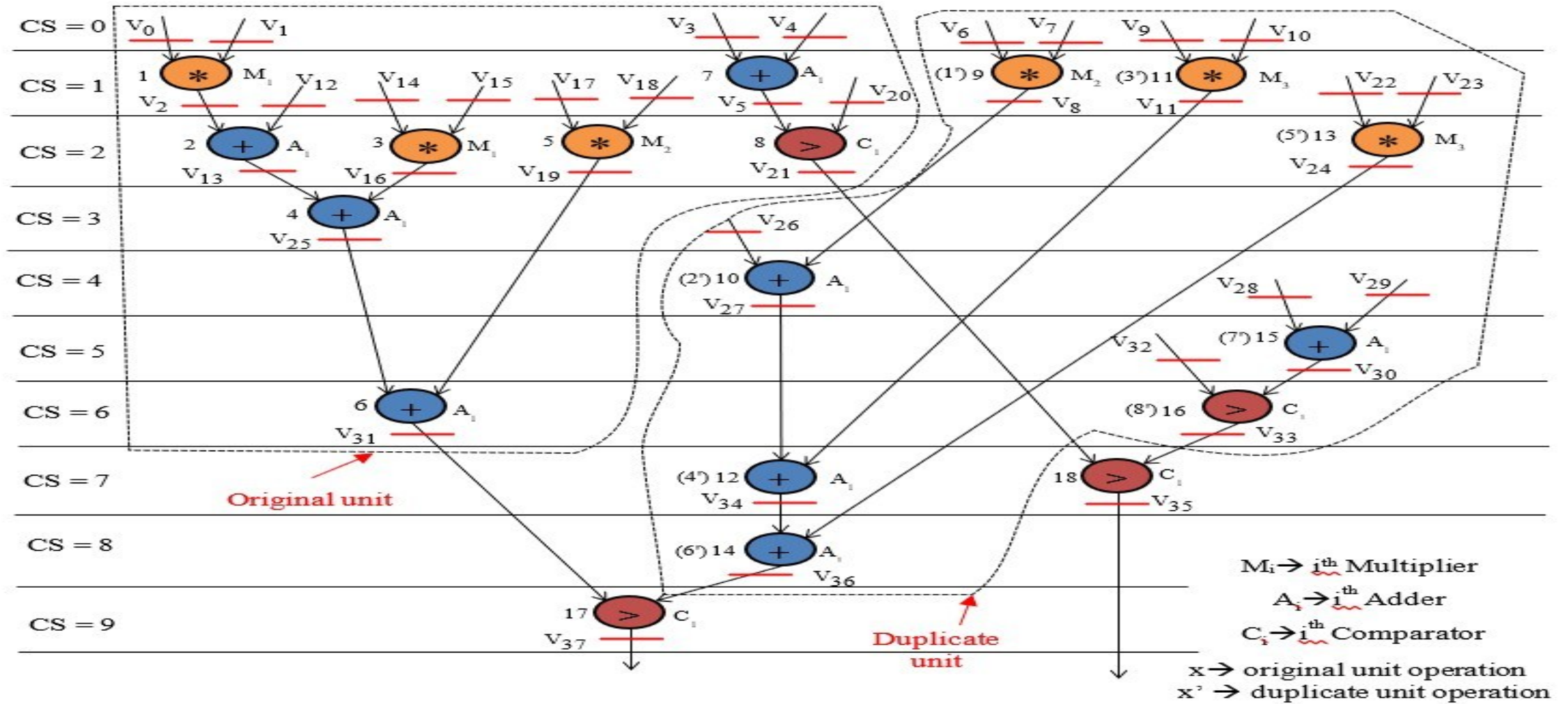
- A motivational example is described to demonstrate working of the approach for producing digital signature implanted SDFG^{DMR} of fault secured finite impulse response (FIR) core.
- The equation for loop based FIR filter is shown below:

$$y[n] = \sum_{i=0}^8 h[i] * x[n - i] \quad (1)$$

where, $y[n]$ is the output of FIR filter, $x[n-i]$ denotes the input impulse, $h[i]$ is constant value and variable 'i' iterates from 0 to 8.

- The loop based CDFG representing the above eq. (1) and UF value are given as input to the phase-1 of presented approach.
 - For this demonstration, UF value = 3 and resource configuration: 3(*), 1(+), 1(>) are considered.
 - The given CDFG is unrolled based on input UF value and passed to phase-2 where transient fault secured FIR design is created based on fault strength ($K_c=2$).
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(a) Constructing a Transient Fault Secured DSP design:

Fig. Transient fault secured design (SDFG^{DMR}) of FIR filter ($UF=3$)

(b) Generating the Digital Signature

- Phase-3 begins with generating a bitstream using the scheduling information of $SDFG^{DMR}$ according to the encoding rule-2.
 - The bitstream generated in this demonstration for FIR filter is 001110000001100001.
 - A bitstream digest is formed by providing the bitstream generated in the preceding step as input to SHA-512.
 - Thereafter, the 512-bit buffer is initialized and main hash computation is performed.
 - The hashed bitstream output obtained after performing SHA-512 is 10000100010111101001110.....111001010100010011 (512 bits).
 - This 512-bit hashed bitstream is divided into 64 blocks of equal length and each block is converted to equivalent decimal value.
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Cont.

- The first 10 blocks and their equivalent decimal value are shown.
- Next, these decimal values are encrypted using RSA encryption.
- This encrypted hashed bitstream is termed as digital signature.
- For this demonstration, a digital signature size of $S=15$ is considered. The digital signature 101100110010000 is embedded in the $SDFG^{DMR}$ in phase-4.

Hashed bitstream blocks	Decimal Equivalent Input	Encrypted Decimal Output	Binary Equivalent of output
10000100	132	89	1011001
01011110	94	289	100100001
10011101	157	222	11011110
01100000	96	8	1000
11001100	204	113	1110001
00010100	20	136	10001000
10111110	190	187	10111011
01101111	111	84	1010100
10010010	146	81	1010001
01110100	116	116	1110100

Bits in digital signature (S=15)	Corresponding additional edges
1	$\langle V_2, V_4 \rangle$
0	$\langle V_2, V_3 \rangle$
1	$\langle V_2, V_6 \rangle$
1	$\langle V_2, V_8 \rangle$
0	$\langle V_2, V_5 \rangle$
0	$\langle V_2, V_7 \rangle$
1	$\langle V_2, V_{10} \rangle$
1	$\langle V_2, V_{12} \rangle$
0	$\langle V_2, V_{11} \rangle$
0	$\langle V_2, V_{13} \rangle$
1	$\langle V_2, V_{14} \rangle$
0	$\langle V_2, V_{17} \rangle$
0	$\langle V_2, V_{19} \rangle$
0	$\langle V_2, V_{23} \rangle$
0	$\langle V_2, V_{29} \rangle$

Cont.

- Next, the digital signature bits are mapped to the extra v_0 edges that are to be added in the CIG to impose the watermarking constraints.

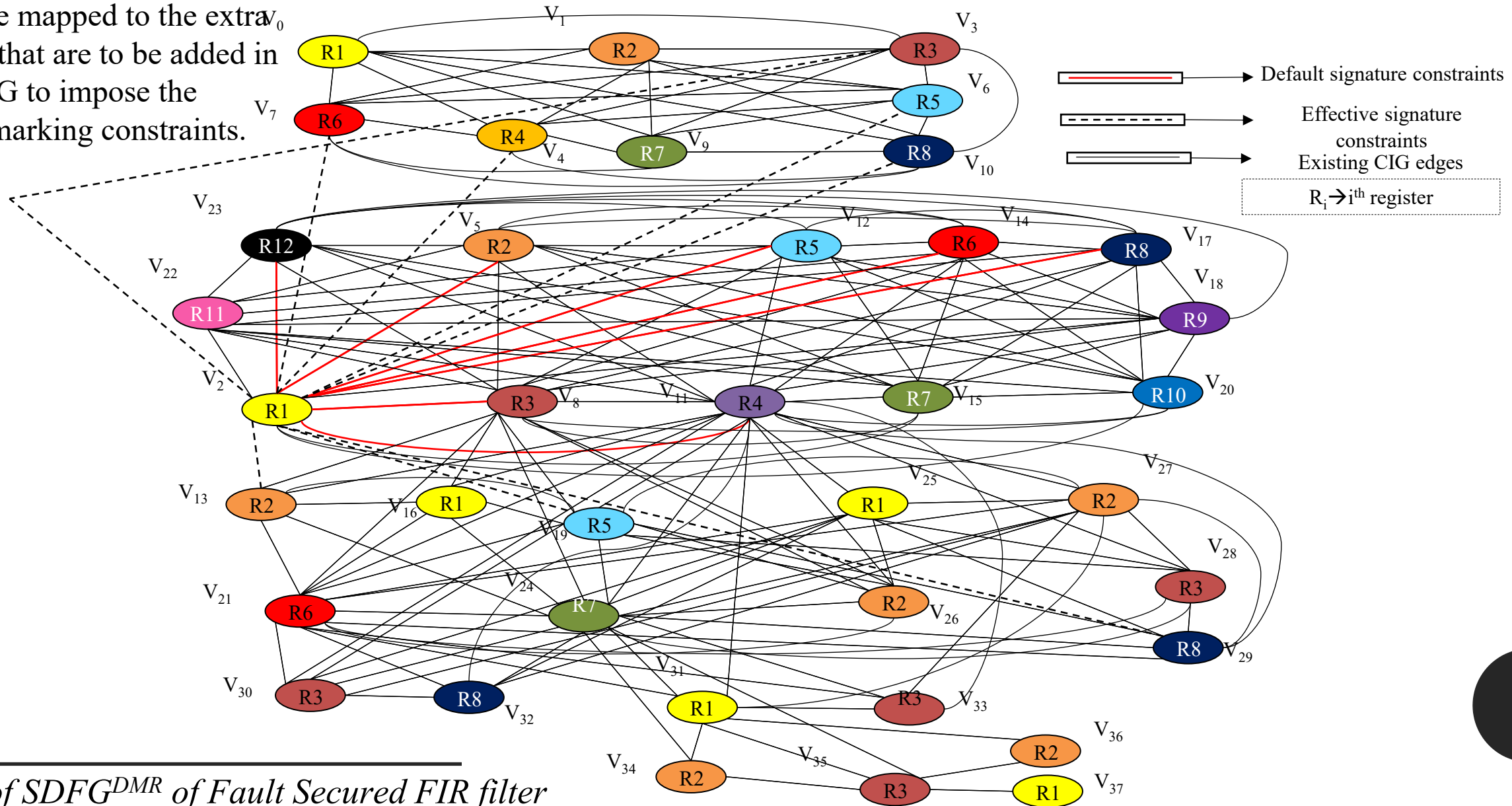


Fig. CIG of $SDFG^{DMR}$ of Fault Secured FIR filter ($UF=3$) after embedding digital signature ($S = 15$)

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Register allocation table of Fault Secured FIR LOOP (UF=3) after embedding digital signature
(S = 15)

Control Step	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12
0	V ₀	V ₁	V ₃	V ₄	V ₆	V ₇	V ₉	V ₁₀	-	-	-	-
1	V ₂	V ₅	V ₈	V ₁₁	V ₁₂	V ₁₄	V ₁₅	V ₁₇	V ₁₈	V ₂₀	V ₂₂	V ₂₃
2	V ₁₆	V ₁₃	V ₈	V ₁₁	V ₁₉	V ₂₁	V ₂₄	-	-	-	-	-
3	V ₂₅	V ₂₆	V ₈	V ₁₁	V ₁₉	V ₂₁	V ₂₄	-	-	-	-	-
4	V ₂₅	V ₂₇	V ₂₈	V ₁₁	V ₁₉	V ₂₁	V ₂₄	V ₂₉	-	-	-	-
5	V ₂₅	V ₂₇	V ₃₀	V ₁₁	V ₁₉	V ₂₁	V ₂₄	V ₃₂	-	-	-	-
6	V ₃₁	V ₂₇	V ₃₃	V ₁₁	-	V ₂₁	V ₂₄	-	-	-	-	-
7	V ₃₁	V ₃₄	V ₃₅	-	-	-	V ₂₄	-	-	-	-	-
8	V ₃₁	V ₃₆	V ₃₅	-	-	-	-	-	-	-	-	-
9	V ₃₇	-	V ₃₅	-	-	-	-	-	-	-	-	-

- Once the register allocation table is modified, the RTL datapath generated earlier is modified to finally obtain a watermarked transient fault secured FIR filter.

Analysis on Case Studies – DSP cores

❑ The metrics used to evaluate the digital signature based watermark are as follows:

- (i) Design cost of the presented approach of protecting fault secured DSP IP cores using digital signature based watermark
 - (ii) Robustness of the presented approach
 - (iii) Impact on storage overhead on increasing digital signature size
 - (iv) Comparison of robustness and storage overhead with contemporary approach
 - (v) Comparison with baseline design in terms of storage overhead.
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Cont.

1. Security Analysis

- Probability of coincidence (P_c) adopted from (Koushanfar et al., 2005) is used as the security metric for the presented approach which indicates proof of ownership.
- The equation to compute P_c value is given as:

$$P_c = \left(1 - \frac{1}{c}\right)^S \quad (2)$$

- Where, c = number of registers used in the original (baseline) design, S = digital signature size.
 - P_c values for the presented approach for five different digital signature sizes and seven fault secured DSP applications are reported in Table below:
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Cont.

Variation of Probability of Coincidence for different digital signature sizes for digital signature based watermark approach

Benchmarks	c	Size of Digital signature (S)				
		S = 15	S = 30	S = 60	S = 120	S = 240
		P _c	P _c	P _c	P _c	P _c
BPF	12	0.2711	0.0735	5.4036x10 ⁻³	2.9199x10 ⁻⁵	8.526x10 ⁻¹⁰
JPEG SAMPLE	22	0.4977	0.2477	0.0613	3.7635x10 ⁻³	1.4164x10 ⁻⁵
MESA FEEDBACK POINTS	33	0.6303	0.3973	0.1578	0.0249	6.2037x10 ⁻⁴
MESA INTERPOLATE AUX	77	0.8219	0.6756	0.4564	0.2083	0.0434
IDCT	16	0.3798	0.1443	0.0208	4.3307x10 ⁻⁴	1.8755x10 ⁻⁷
FIR LOOP (UF=2)	9	0.1709	0.0292	8.5280x10 ⁻⁴	7.2727x10 ⁻⁷	5.2893x10 ⁻¹³
FIR LOOP (UF=4)	12	0.2711	0.0735	5.4036x10 ⁻³	2.9199x10 ⁻⁵	8.526x10 ⁻¹⁰

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- Further, the percentage reduction in P_c value for the presented approach compared to (Sengupta and Bhaudauria, 2016) is highlighted in Fig. below:

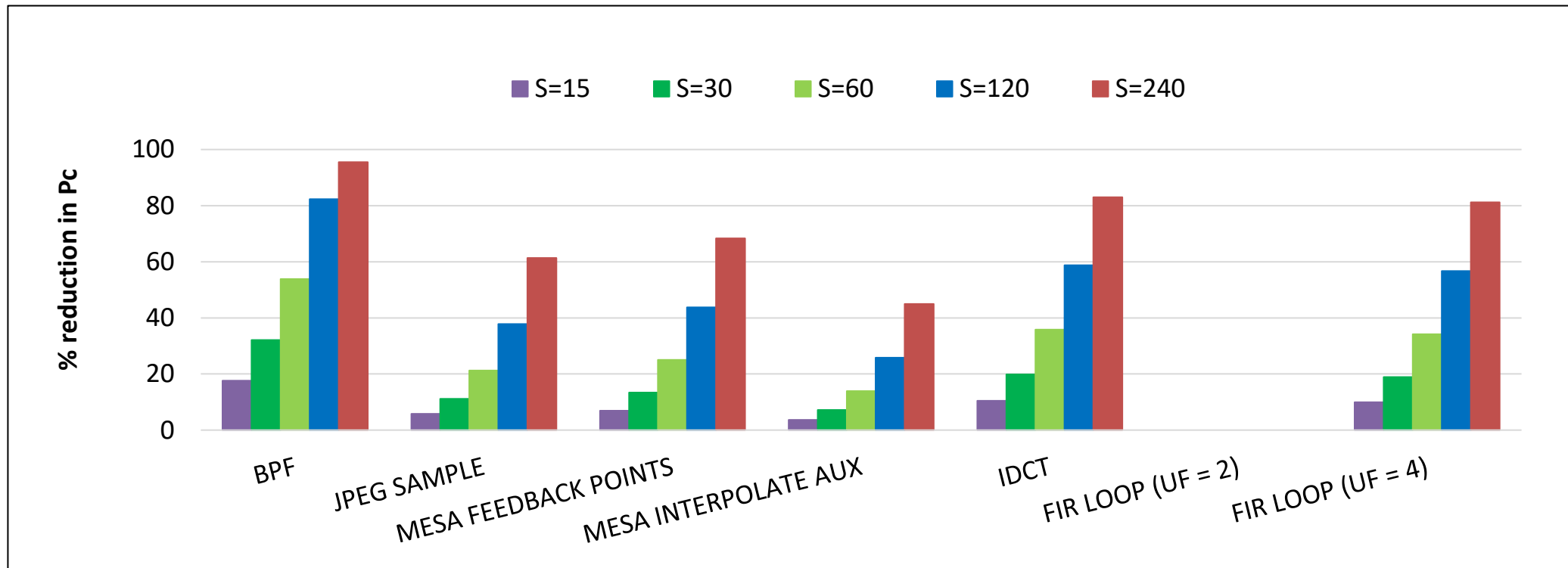


Fig. % reduction in P_c (i.e. enhancement of proof of ownership) of digital signature based watermark approach compared to (Sengupta and Bhaudauria, 2016)

Design Cost Analysis

- Embedding cost of presented approach is evaluated in terms of design solution area and latency.

Function used to compute embedding cost is as follows:

$$C_f(X_i) = \phi_1 \frac{L_T}{L_{max}} + \phi_2 \frac{A_T}{A_{max}} \quad (3)$$

- where, $C_f(X_i)$ is the cost of the embedding watermark in transient fault secured design solution for resource configuration input X_i ;
 - A_T and L_T denote area and latency of a fault secured watermarked design solution respectively;
 - A_{max} and L_{max} are maximum possible area and latency of a design solution and
 - ϕ_1 and ϕ_2 are user chosen weights both equal to 0.5 for equal priority of area and delay.
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- ❑ The design cost values computed after embedding digital signature of size 60 for different fault secured DSP cores are reported in Table below:

Results for digital signature ($S = 60$) solution through the digital signature-based watermark approach

DSP applications	Presented Solution	A_T (μm^2)	L_T (ps)	Design Cost
BPF	3(+), 4(*), 1(>)	372.0808	15000	0.3139
JPEG SAMPLE	8(+), 2(*), 1(>)	322.5352	14000	0.2849
IDCT	8(+), 4(*), 3(>)	475.3	11000	0.1709
FIR LOOP (UF =2)	1(+), 2(*), 1(>)	180.1912	8000	0.5656
FIR LOOP (UF =4)	1(+), 3(*), 1(>)	258.048	11000	0.5165
MESA FEEDBACK POINTS	6(+), 8(*), 4(>)	756.1544	9000	0.2415
MESA INTERPOLATE AUX	6(+), 16(*), 3(>)	1391.4928	20000	0.2140

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- ❑ The comparison of number of registers required post-embedding digital signature, for presented approach and (Sengupta and Bhaudauria, 2016) for different fault secured DSP cores and w.r.t different digital signature sizes (watermark strengths)

DSP applications	# of registers required in presented approach					# of registers required in (Sengupta and Bhaudauria, 2016)				
	S = 15	S = 30	S = 60	S = 120	S = 240	S = 15	S = 30	S = 60	S = 120	S = 240
BPF	12	12	13	13	15	14	15	16	17	17
JPEG SAMPLE	22	22	22	22	22	24	24	25	25	26
MESA FEEDBACK POINTS	33	33	33	33	33	39	39	39	40	40
MESA INTERPOLATE AUX	77	77	77	77	78	95	95	95	95	96
IDCT	16	16	16	16	16	18	18	18	19	19
FIR_LOOP (UF = 2)	9	9	9	NA	NA	9	10	10	NA	NA
FIR LOOP (UF = 4)	12	12	12	13	NA	13	13	14	14	NA

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❑ *Comparison with Baseline Design in terms of Storage Hardware*

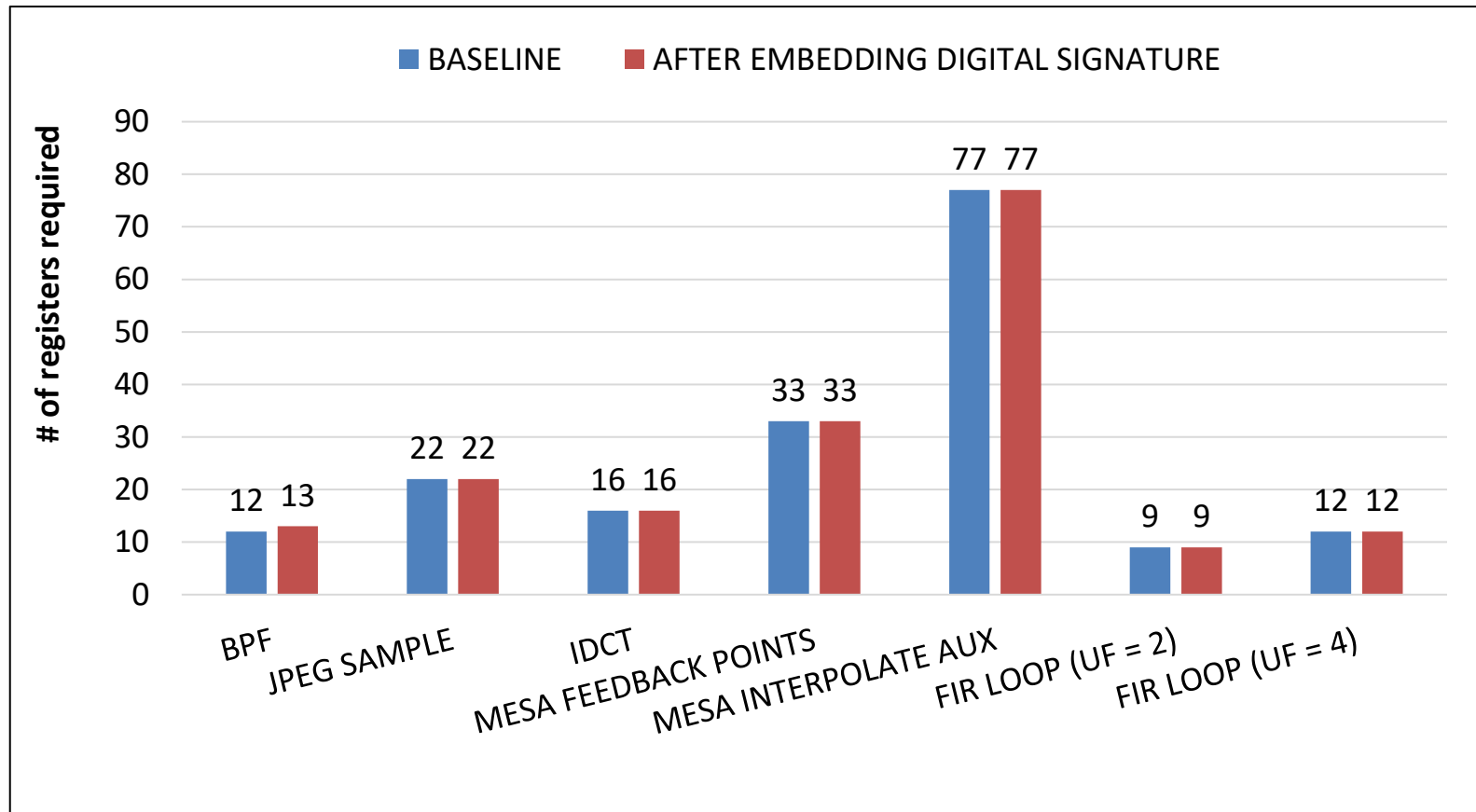


Fig. *Comparison of # of register hardware used in baseline design and after embedding digital signature ($S = 60$)*

Conclusion

- ✓ A multi-level encoding and encrypted-hash based digital signature for protection of fault secured DSP cores against IP piracy and false claim of ownership has been presented.
 - ✓ The digital signature embedding approach achieved stronger proof of ownership while incurring less register hardware overhead compared to contemporary works.
 - ✓ The analysis on case studies reveal that digital signature based watermark on average obtained an reduction of Pc and storage hardware of 26.27% and 14.35 % respectively compared to (Sengupta and Bhaudauria, 2016).
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Thank You

